

Project 1: Resource Monitoring Dashboard Using Python

Implementation Assignment – 1

Title: Resource Monitoring Dashboard for Virtual Machine Resources

Objective:

Develop a Python-based application to monitor and display the real-time utilization of system resources, including CPU, memory, disk, and network usage.

Problem Statement

Develop a Python application that continuously monitors the system resources of a computer or virtual machine. The application should display the following resource utilization statistics:

- CPU Usage (%)
- Memory Usage (%)
- Disk Usage (%)
- Network Statistics (Bytes Sent and Bytes Received)

Students may implement either:

- a command-line interface (CLI), or
- a graphical dashboard using Python libraries.

The application should refresh the resource information at regular intervals and present the current utilization in a clear format.

Suggested Tools

- Python 3.x
- psutil
- matplotlib (optional)
- tkinter or streamlit (optional)

Deliverables

The report should include:

1. Objective
2. Software and Libraries Used
3. Installation Commands
4. Source Code
5. Explanation of Important Functions
6. Execution Commands
7. Screenshots of Program Execution
8. Resource Utilization Results
9. Observations

10. Conclusion